Scalar AGY Determinants Survive: Infinite Oscillator Twists Do Not

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August 10, 2026

Abstract

We establish a same-domain separation between scalar analytic transfer theory and an unsmoothed infinite oscillator twist for a countable Avila–Gouëzel–Yoccoz (AGY) return map. Every inverse branch contains one fixed strictly positive Rauzy prefix. Its projective action maps the closure of a canonical complex positive cone into the cone interior, producing a single bounded domain in complex dimension three on which all countably many branches have a common compact image. The raw scalar weights are holomorphic and summable throughout the AGY half-plane $\Re s > -\sigma_0$. The resulting Bergman transfer operator belongs to an exponential singular-value class and is trace class. For a chronological word matrix $A \in \mathrm{SL}(4, \mathbb{Z})$, with Perron root λ and $\chi_A(t) = \det(tI - A)$, its ordinary trace atom is exactly

$$\frac{\lambda^{-(s+1)}}{\chi_A'(\lambda)}.$$

The exponent follows from a cancellation between the four-letter Jacobian and the three-dimensional projective derivative. On the same Bergman domain, adjoining the literal metaplectic operators on $L^2(\mathbb{R}^2)$ produces a bounded but noncompact operator: constants and evaluation expose an absolutely summable family of distinct oscillator atoms with positive essential norm. An independently replayed length-128 branch gives the exact normalizer 15076979616018/8999921. Thus the scalar dynamical determinant survives, whereas the ordinary infinite-fibre Fredholm proposal does not. The result does not supply a self-adjoint Hilbert–Pólya operator or a Riemann-zero correspondence; it instead identifies finite Weil fibres as the next well-posed arithmetic test.

1 Introduction

Analytic contraction of a scalar base does not compactify an infinite-dimensional unitary fibre. This elementary warning becomes a decisive operator theorem for the countable induced Rauzy system constructed by Avila, Gouëzel, and Yoccoz (AGY) [1]. The scalar system has the geometry required by holomorphic transfer theory, while the literal oscillator twist remains noncompact on the very same complex domain.

The operator-space question is subtle because the obvious real-variable localizer disappears after complexification. In a bounded C^1 space, a bump supported in one inverse-branch cylinder can isolate one unitary fibre operator. A nonzero holomorphic function cannot have that support. It is therefore not enough to transfer a real noncompactness proof to a Bergman or Hardy space. Conversely, scalar nuclearity theorems for compactly contained holomorphic branches do not automatically apply to operator-valued weights; the compactness criteria of Bonet et al. [4] likewise retain an explicit fibre condition.

We resolve both sides without changing the dynamics. AGY's branch grammar makes each return either the fixed strongly positive path γ_* or a minimal path $\gamma_*\gamma_0\gamma_*$ [1, Section 4.1.3 and Lemma 4.4, pp. 162, 165]. Later-on-the-left Rauzy multiplication and transposition then give our factorization

$$A_\gamma = B_\gamma^T = PC_\gamma, \qquad P > 0, \quad C_\gamma \geq 0.$$

Table 1: The same AGY branches and the same complex domain give opposite ordinary determinant outcomes. "Unavailable" means unavailable as an ordinary Hilbert-space trace; distributional characters are a separate framework.

Property	Scalar $A^2(\Omega)$	Oscillator $A^2(\Omega; L^2(\mathbb{R}^2))$
Branch series	Trace-norm summable	Operator-norm summable
Compactness	Exponential singular-value class	Noncompact
Ordinary trace	Defined	Unavailable
Periodic atom	$\lambda^{-(s+1)}/\chi'(\lambda)$	No ordinary metaplectic factor
Fredholm determinant	$\det(I - uL_s)$ entire in u	No ordinary nuclear determinant
Decisive mechanism	Positive-prefix compact containment	Evaluation slice exposes unitary atoms

The fixed map p_P sends the closure of a canonical complex cone strictly inside that cone, whereas every p_{C_γ} preserves it. An enlarged intermediate domain Ω therefore satisfies

$$\bigcup_{\gamma} h_\gamma(\Omega) \Subset \Omega$$

for the complete countable return family. The real AGY exponential-tail estimate then controls the complex weights through one right-half-plane logarithm. The Bergman results of Bandtlow and Jenkinson [2, 3] yield a legitimate scalar Fredholm determinant and an ordinary fixed-point trace formula.

The scalar periodic atoms simplify further. A later-on-the-left return word has a positive integral matrix $A_{\boldsymbol{\gamma}}$, a simple Perron root $\lambda_{\boldsymbol{\gamma}}$, and a monic characteristic polynomial $\chi_{\boldsymbol{\gamma}}$. Normalizers telescope to the Perron root, while the projective derivative is the quotient action of $A_{\boldsymbol{\gamma}}/\lambda_{\boldsymbol{\gamma}}$. Hence

$$\operatorname{tr} T_{s,\boldsymbol{\gamma}} = \frac{\lambda_{\boldsymbol{\gamma}}^{-(s+1)}}{\chi_{\boldsymbol{\gamma}}'(\lambda_{\boldsymbol{\gamma}})}. \tag{1}$$

This formula is arithmetic in a precise but limited sense: the Perron root is an algebraic unit attached to a chronological integer matrix. No prime-orbit law, automorphic interpretation, or field-discriminant identity is inferred.

The infinite fibre is tested after the scalar construction is complete. Let U_γ be the metaplectic operator obtained by composing the declared edge lifts in true time order on $\mathscr{F} = L^2(\mathbb{R}^2)$; standard phase-space conventions are recalled from Folland [5]. Constants and evaluation at a real interior point compress the vector-valued Bergman operator to

$$\sum_{\gamma} w_{s,\gamma}(x_0) U_\gamma.$$

The projected symplectic atoms are pairwise distinct, and coherent-state separation gives

$$\|\mathcal{L}_s^{\mathrm{Mp}}\|_{\mathrm{ess}} \geq \frac{\left(\sum_\gamma |w_{s,\gamma}(x_0)|^2\right)^{1/2}}{\|E_{x_0}\|\,\|J\|} > 0. \tag{2}$$

No branch-supported holomorphic function is used.

The contributions are therefore concrete and falsifiable:

1. We construct one bounded complex domain for every inverse branch of the countable AGY return map using only the fixed positive prefix.

2. We prove scalar Bergman trace class, trace-norm word interchange, and the exact Perron–characteristic formula (1).

3. We prove the same-domain noncompactness bound (2) for the unsmoothed oscillator twist.

4. We provide exact and independently replayed arithmetic for a length-128 source branch, a two-return matrix-order sentinel, and a noncyclic three-return spectral-order sentinel.

The paper separates inherited inputs from new deductions. AGY supply the branches, real summability, and positive-path grammar [1]; the fixed-left-factor identity is our chronological algebraic deduction. Prior project theorems supply all-length branch matrix injectivity and the abstract discrete-metaplectic atom estimate; Appendix B restates the needed form. The common complex domain, scalar trace theorem, Perron reduction, and same-domain evaluation application are the present results. Section 2 constructs the domain, Sections 3 and 4 establish the scalar determinant, and Section 5 proves the obstruction. Exact certificates and the Hilbert–Pólya assessment follow in Sections 6 and 7.

2 Positive-prefix complexification

The common complex domain is our algebraic consequence of the published AGY branch grammar and Rauzy chronology, not a formal complexification of real contraction. We keep the alphabet size

$$d = 4$$

separate from the projective complex dimension $q = d - 1 = 3$. Put

$$\ell(z) = \mathbf{1}^T z, \qquad H_{\mathbb{C}} = \{z \in \mathbb{C}^d : \ell(z) = 1\} \simeq \mathbb{C}^q, \qquad p_M(z) = \frac{Mz}{\ell(Mz)}.$$

For the fixed neat strongly positive return path γ_*, let

$$P = B_{\gamma_*}^T > 0.$$

AGY's Section 4.1.3 and Lemma 4.4 make each return either γ_* or a minimal path $\gamma_*\gamma_0\gamma_*$[1, pp. 162, 165]. Later Rauzy arrows multiply on the left, so transposition places $B_{\gamma_*}^T$ on the left. We therefore deduce

$$A_\gamma := B_\gamma^T = PC_\gamma, \qquad C_\gamma \geq 0, \qquad h_\gamma = p_{A_\gamma} = p_P \circ p_{C_\gamma}. \tag{3}$$

Every C_γ is invertible and therefore has no zero row or column. The direct real formulas are

$$q_\gamma(x) = \ell(A_\gamma x), \qquad r_\gamma(x) = \log q_\gamma(x), \qquad j_\gamma(x) = q_\gamma(x)^{-d}. \tag{4}$$

The sign and exponent in (4) are fixed by the published inverse-Jacobian formula [1].

Define the canonical complex section of the positive cone by

$$\mathcal{D} = \{z \in H_{\mathbb{C}} : \Re(z_i \overline{z_j}) > 0 \text{ for every } i, j\}. \tag{5}$$

Unlike a coordinatewise right half-plane alone, $\mathcal{D}$ is stable under every nonnegative matrix relevant to (3).

Lemma 2.1 (Complex-cone geometry)**.** *The set $\mathcal{D}$ is a bounded connected domain containing the real positive simplex. If $C \geq 0$ is invertible, then*

$$p_C(\mathcal{D}) \subset \mathcal{D}.$$

If $P > 0$, then

$$p_P(\overline{\mathcal{D}}) \Subset \mathcal{D}.$$

Proof. For $z \in \mathcal{D}$, the affine constraint gives

$$\Re z_i = \Re\left(z_i \overline{\sum_j z_j} \right) = \sum_j \Re(z_i \overline{z_j}) > 0.$$

On $\overline{\mathcal{D}}$, all summands below are nonnegative and

$$1 = |\ell(z)|^2 = \sum_{i,j} \Re(z_i \overline{z_j}) \geq |z_k|^2.$$

Thus $\overline{\mathcal{D}}$ is bounded. If $b = d^{-1}\mathbf{1}$, the segment $(1-t)z + tb$ remains in $\mathcal{D}$: expanding each pairwise real product gives a positive combination of $\Re(z_i \bar{z}_j)$, $\Re z_i$, $\Re z_j$, and 1. Hence $\mathcal{D}$ is connected.

For $C \geq 0$, the absence of zero rows implies

$$\Re\big((Cz)_i \overline{(Cz)_j}\big) = \sum_{k,l} C_{ik} C_{jl} \Re(z_k \overline{z_l}) > 0.$$

The coordinate sum of Cz cannot vanish, and division by that common complex scalar preserves all signs. This proves $p_C(\mathcal{D}) \subset \mathcal{D}$.

Let now $P > 0$ and $z \in \overline{\mathcal{D}}$. The same double sum is nonnegative, while its diagonal part

$$\sum_k P_{ik} P_{jk} |z_k|^2$$

is strictly positive. Therefore $p_P(z) \in \mathcal{D}$. The denominator stays nonzero on the compact set $\overline{\mathcal{D}}$, so continuity yields the compact containment $p_P(\overline{\mathcal{D}}) \Subset \mathcal{D}$. $\square$

Theorem 2.2 (One domain for the countable AGY family)**.** *There is a bounded connected domain $\Omega \subset H_{\mathbb{C}}$ such that*

$$p_P(\overline{\mathcal{D}}) \Subset \Omega \Subset \mathcal{D} \tag{6}$$

and

$$\bigcup_{\gamma \in \Gamma} h_\gamma(\Omega) \subset p_P(\mathcal{D}) \Subset \Omega. \tag{7}$$

Proof. The compact set $K_P = p_P(\overline{\mathcal{D}})$ is connected. Choose a small connected Euclidean neighborhood Ω of K_P whose closure lies in $\mathcal{D}$. By (3) and Lemma 2.1,

$$h_\gamma(\Omega) = p_P p_{C_\gamma}(\Omega) \subset p_P(\mathcal{D}) \subset K_P \Subset \Omega$$

for every branch. The gap is independent of γ, including limits of the countable branch family. $\square$

The enlargement in (6) is essential. The natural domain $p_P(\mathcal{D})$ contains the real AGY base, but the real branch images partition that base modulo null sets and approach its boundary. They cannot all lie in one compact subset of the same natural domain. The intermediate Ω contains the closure of the natural image and places that closure strictly inside a larger holomorphic domain.

Real Hilbert contraction alone would not prove Theorem 2.2: a family of rational functions can contract one real interval uniformly while poles approach it from imaginary directions. Our proof avoids that failure because the matrices have the fixed positive left factor deduced above and no branch-dependent poles on $\mathcal{D}$. An explicit one-dimensional counterexample and a matrix-normalization audit appear in Appendix A.

3 The scalar holomorphic transfer operator

The raw scalar weight is

$$w_{s,\gamma}(x) = e^{-sr_\gamma(x)} j_\gamma(x) = q_\gamma(x)^{-(s+d)}.$$

We next verify the two countable hypotheses that cannot be inferred from compact containment alone: one logarithm for all branches and an ℓ^1 weight bound.

Lemma 3.1 (Common logarithm and complex branch sum). *For every $s \in \{s \in \mathbb{C} : \Re s > -\sigma_0\}$, all weights extend holomorphically to Ω by*

$$w_{s,\gamma}(z) = \exp(-(s+d)\operatorname{Log} \ell(A_\gamma z)), \tag{8}$$

where Log *is the principal logarithm. Moreover,*

$$\sum_{\gamma\in\Gamma} \|w_{s,\gamma}\|_{H^\infty(\Omega)} < \infty. \tag{9}$$

The convergence is uniform when s ranges over a compact subset of $\{s \in \mathbb{C} : \Re s > -\sigma_0\}$.

Proof. Write

$$q_\gamma(z) = \ell(A_\gamma z) = c_\gamma \cdot z, \qquad c_{\gamma,j} = \sum_i (A_\gamma)_{ij} > 0.$$

Because $\overline{\Omega} \Subset \mathcal{D}$, there are constants

$$\delta_\Omega = \min_{z\in\overline{\Omega},i} \Re z_i > 0, \qquad M_\Omega = \max_{z\in\overline{\Omega},i} |z_i| < \infty.$$

It follows that

$$\Re q_\gamma(z) \geq \delta_\Omega \sum_j c_{\gamma,j} > 0.$$

Thus every denominator lies in the same right half-plane and (8) is unambiguous.

Fix a comparison point $x_0 \in \Delta \subset p_P(\mathcal{D}) \subset \Omega$ and put $m_0 = \min_i (x_0)_i$, $M_0 = \max_i (x_0)_i$. Uniformly in z and γ,

$$\frac{\delta_\Omega}{M_0} \leq \frac{|q_\gamma(z)|}{q_\gamma(x_0)} \leq \frac{M_\Omega}{m_0}, \qquad |\arg q_\gamma(z)| < \frac{\pi}{2}. \tag{10}$$

For fixed s, taking moduli in (8) and using both sides of (10) gives

$$\|w_{s,\gamma}\|_\infty \leq C_{\Omega,s} q_\gamma(x_0)^{-(d+\Re s)}.$$

The AGY exponential-tail and bounded-distortion estimates imply the real branch sum

$$\sum_\gamma \sup_{x\in\Delta} q_\gamma(x)^{-(d+\Re s)} < \infty, \qquad \Re s > -\sigma_0; \tag{11}$$

see the source construction in Avila et al. [1]. This proves (9). For local uniformity, let $K \Subset \{s \in \mathbb{C} : \Re s > -\sigma_0\}$ and choose $-\sigma_0 < a_0 < \min_{s\in K} \Re s$. The inherited return-time convention gives $q_\gamma(x_0) = e^{r_\gamma(x_0)} \geq 2$. The sector factor $e^{|\Im s|\pi/2}$ and the comparison powers in (10) are bounded on K, and the inequality $q_\gamma(x_0) \geq 2$ therefore gives the single majorant

$$\sup_{s\in K} \|w_{s,\gamma}\|_\infty \leq C_K q_\gamma(x_0)^{-(d+a_0)}.$$

Its sum converges by (11) at a_0. This proves local uniformity; no bound uniform as $|\Im s| \to \infty$ is asserted. $\square$

Let $A^2(\Omega)$ denote the Bergman space with respect to six-dimensional Euclidean volume on $H_{\mathbb{C}} \simeq \mathbb{C}^3$. Define

$$(L_s f)(z) = \sum_{\gamma \in \Gamma} w_{s,\gamma}(z) f(h_\gamma z). \tag{12}$$

Theorem 3.2 (Scalar Bergman trace class)**.** *For every $s \in \{s \in \mathbb{C} : \Re s > -\sigma_0\}$, the operator L_s belongs to an exponential singular-value class $E(c, 1/3)$, for a domain-dependent $c > 0$. In particular, L_s is trace class and*

$$D(s,u) = \det_{A^2(\Omega)} (I - uL_s) \tag{13}$$

is entire in u. The map $s \mapsto L_s$ is locally holomorphic in trace norm on $\{s \in \mathbb{C} : \Re s > -\sigma_0\}$, so D is jointly holomorphic on $\{s \in \mathbb{C} : \Re s > -\sigma_0\} \times \mathbb{C}$.

Proof. By Theorem 2.2 and lemma 3.1, the pairs $(h_\gamma, w_{s,\gamma})$ form a countable holomorphic map-weight system with one common compact image and a summable weight majorant. The Bergman embedding theorem of Bandtlow and Jenkinson [2] places the associated transfer operator in $E(c, 1/q)$, here $q = 3$. Stretched-exponential singular-value decay implies trace class.

For the parameter statement, choose an intermediate domain

$$\overline{\bigcup_\gamma h_\gamma(\Omega)} \subset \widetilde{\Omega} \Subset \Omega. \tag{14}$$

Restriction $J_\Omega : A^2(\Omega) \to A^2(\widetilde{\Omega})$ is trace class. Each branch factors as

$$T_{s,\gamma} = \widetilde{T}_{s,\gamma} J_\Omega, \qquad \|\widetilde{T}_{s,\gamma}\| \le C_\Omega \|w_{s,\gamma}\|_\infty.$$

Consequently

$$\sum_\gamma \|T_{s,\gamma}\|_1 \le C_\Omega \|J_\Omega\|_1 \sum_\gamma \|w_{s,\gamma}\|_\infty < \infty, \tag{15}$$

locally uniformly in s. The Weierstrass theorem for Banach-valued holomorphic series gives trace-norm holomorphy, and the standard Fredholm determinant then gives the joint conclusion. These conclusions also follow from the favorable-space nuclear framework of Bandtlow and Jenkinson [3]. □

The theorem concerns the raw transfer operator in (12). AGY prove that the invariant density is bounded and C^1, but not that it extends holomorphically to Ω. We do not insert that density into the holomorphic operator. At this point the scalar determinant is legal; the infinite oscillator fibre has not yet been added.

4 Perron arithmetic of chronological trace atoms

The order convention matters before any spectral reduction. Let $\boldsymbol{\beta} = (\beta_1, \ldots, \beta_n)$ list AGY returns in forward time, from earliest to latest. Rauzy matrices multiply later on the left:

$$B_{\boldsymbol{\beta}} = B_{\beta_n} \cdots B_{\beta_1}, \qquad A_{\boldsymbol{\beta}} = B_{\boldsymbol{\beta}}^T = A_{\beta_1} \cdots A_{\beta_n}. \tag{16}$$

The inverse projective map and the corresponding term in L_s^n are

$$h_{\boldsymbol{\beta}} = h_{\beta_1} \circ \cdots \circ h_{\beta_n} = p_{A_{\boldsymbol{\beta}}}, \qquad T_{s,\beta_n} \cdots T_{s,\beta_1}. \tag{17}$$

Equivalently, if $(\gamma_1, \ldots, \gamma_n)$ indexes the operator product $T_{s,\gamma_1} \cdots T_{s,\gamma_n}$, then $A = A_{\gamma_n} \cdots A_{\gamma_1}$. This equivalence prevents a transpose from silently reversing time.

Every $A_{\boldsymbol{\beta}}$ is a strictly positive matrix in $\mathrm{SL}(d, \mathbb{Z})$. Let $\lambda_{\boldsymbol{\beta}} > 0$ be its Perron root and put

$$\chi_{\boldsymbol{\beta}}(t) = \det(tI - A_{\boldsymbol{\beta}}) \in \mathbb{Z}[t]. \tag{18}$$

Theorem 4.1 (Perron–characteristic trace formula). *The map $h_{\boldsymbol\beta}$ has one fixed point in Ω, the normalized positive Perron vector of $A_{\boldsymbol\beta}$, and*

$$\boxed{\operatorname{tr}\bigl(T_{s,\beta_n}\cdots T_{s,\beta_1}\bigr) = \frac{\lambda_{\boldsymbol\beta}^{-(s+1)}}{\chi'_{\boldsymbol\beta}(\lambda_{\boldsymbol\beta})}.} \tag{19}$$

For every fixed n,

$$\boxed{\operatorname{tr} L_s^n = \sum_{\boldsymbol\beta\in\Gamma^n} \frac{\lambda_{\boldsymbol\beta}^{-(s+1)}}{\chi'_{\boldsymbol\beta}(\lambda_{\boldsymbol\beta})},} \tag{20}$$

and the sum converges absolutely and locally uniformly for $s \in \{s \in \mathbb{C} : \Re s > -\sigma_0\}$.

Proof. Strict positivity gives a simple Perron root and a positive eigenvector. The corresponding normalized vector belongs to the real AGY base because $A_{\boldsymbol\beta}$ retains the fixed positive left factor after the projective composition. Compact containment gives uniqueness of the fixed point in Ω.

Let x_0 denote that fixed point and follow the inverse periodic history in the order in which the maps in (17) act. At each step write

$$A_{\gamma_k} x_{k-1} = q_k x_k, \qquad q_k = \ell(A_{\gamma_k} x_{k-1}) > 0.$$

Multiplication around the closed history gives

$$A_{\boldsymbol\beta} x_0 = \left(\prod_{k=1}^n q_k\right) x_0, \qquad \prod_{k=1}^n q_k = \lambda_{\boldsymbol\beta}.$$

All intermediate normalizers are positive real numbers, so no principal-log winding occurs. The product of the scalar weights is therefore

$$W_{s,\boldsymbol\beta}(x_0) = \lambda_{\boldsymbol\beta}^{-(s+d)}. \tag{21}$$

Let $V = \ker \ell$, the tangent space of $H_{\mathbb{C}}$. At a normalized Perron vector x, direct differentiation gives

$$Dp_A(x)v = \lambda^{-1}\bigl(Av - x\ell(Av)\bigr), \qquad v \in V. \tag{22}$$

The map $V \to \mathbb{C}^d/\mathbb{C}x$, $v \mapsto [v]$, is an isomorphism. Under this isomorphism (22) is the quotient action of A/λ. Hence, without assuming diagonalizability of the other eigenvalues,

$$\det_{\mathbb{C}}(I - Dp_A(x)) = \frac{\chi'_A(\lambda)}{\lambda^{d-1}}. \tag{23}$$

The fixed-point trace formula for holomorphic weighted composition operators [2, 3] divides (21) by (23). Since the Jacobian exponent is $d = 4$ and the projective dimension is $d - 1 = 3$, the quotient is $\lambda^{-(s+1)}/\chi'_A(\lambda)$.

It remains to justify the countable trace interchange. The branch series converges absolutely in trace norm by (15); therefore

$$\sum_{(\gamma_1,\dots,\gamma_n)\in\Gamma^n} \|T_{s,\gamma_1}\cdots T_{s,\gamma_n}\|_1 \le \left(\sum_{\gamma} \|T_{s,\gamma}\|_1\right)^n < \infty.$$

Taking the continuous Hilbert-space trace term by term proves (20). Local uniformity follows from the compact s-set bound in Lemma 3.1. $\square$

The monic polynomial $\chi_{\boldsymbol{\beta}}$ has constant term one, so its Perron root is an algebraic unit. The denominator $\chi'(\lambda)$ records separation from the other eigenvalues, but it is not automatically a number-field discriminant: the characteristic polynomial may be reducible or nonminimal. For nonintegral complex s, the value $\lambda^{-(s+1)}$ is likewise not asserted to be algebraic.

In the present full-rank four-dimensional model, the word matrix is conjugate to a symplectic matrix, so

$$\chi_{\boldsymbol{\beta}}(t) = t^4 - a_{\boldsymbol{\beta}} t^3 + b_{\boldsymbol{\beta}} t^2 - a_{\boldsymbol{\beta}} t + 1, \qquad a_{\boldsymbol{\beta}}, b_{\boldsymbol{\beta}} \in \mathbb{Z}. \tag{24}$$

The atom depends on this polynomial and its Perron root, but (20) still counts every marked word. Spectral collisions do not quotient the symbolic dynamics.

Finally, the trace-class determinant identity gives

$$-\log D(s,u) = \sum_{n\geq 1} \frac{u^n}{n} \sum_{\boldsymbol{\beta}\in\Gamma^n} \frac{\lambda_{\boldsymbol{\beta}}^{-(s+1)}}{\chi'_{\boldsymbol{\beta}}(\lambda_{\boldsymbol{\beta}})} \tag{25}$$

for sufficiently small $|u|$, locally uniformly on compact subsets of $\{s \in \mathbb{C} : \Re s > -\sigma_0\}$. The Fredholm determinant, not the orbit-log series, gives the entire continuation in u. In particular, evaluation of (25) at $u = 1$ requires an additional convergence argument.

5 The infinite oscillator obstruction

Let $\mathscr{F} = L^2(\mathbb{R}^2)$, and let $U_\gamma = \mu(\widetilde{g}_\gamma)$ be the oscillator representation of the pathwise metaplectic lift. Edge lifts are composed in true time order; the central sign is never averaged. The Weyl representation, metaplectic covariance, and the Schrödinger–Bargmann equivalence used below are standard [5]. Define the literal vector-valued transfer operator by

$$(\mathcal{L}_s^{\mathrm{Mp}} F)(z) = \sum_{\gamma\in\Gamma} w_{s,\gamma}(z) U_\gamma F(h_\gamma z). \tag{26}$$

The noncompactness mechanism applies beyond Bergman space, provided the candidate has two bounded slices.

Theorem 5.1 (Evaluation-slice obstruction)**.** *Let X be a Banach space of $\mathscr{F}$-valued functions on a set containing a real interior point x_0. Assume that*

1. *the constant embedding $J : \mathscr{F} \to X$, $(Jv)(x) = v$, is bounded;*
2. *evaluation $E_{x_0} : X \to \mathscr{F}$ is bounded; and*
3. *(26) defines a bounded operator on X.*

Suppose also that distinct AGY return branches have distinct projected symplectic matrices. Then, for every $s \in \{s \in \mathbb{C} : \Re s > -\sigma_0\}$,

$$\boxed{\|\mathcal{L}_s^{\mathrm{Mp}}\|_{\mathrm{ess}} \geq \frac{\left(\sum_{\gamma\in\Gamma} |w_{s,\gamma}(x_0)|^2\right)^{1/2}}{\|E_{x_0}\|\,\|J\|} > 0.} \tag{27}$$

In particular, $\mathcal{L}_s^{\mathrm{Mp}}$ is noncompact.

Proof. The real branch estimate (11) gives absolute coefficient summability, and evaluation on constants yields the exact full-family compression

$$E_{x_0} \mathcal{L}_s^{\mathrm{Mp}} J = \sum_{\gamma\in\Gamma} w_{s,\gamma}(x_0) U_\gamma. \tag{28}$$

An inherited all-length decoder proves that the fixed-start projected matrices are pairwise distinct. The discrete metaplectic atom theorem, restated and proved in the form needed here in Lemma B.1, gives

$$\|E_{x_0}\mathcal{L}_s^{\mathrm{Mp}}J\|_{\mathrm{ess}} \geq \left(\sum_{\gamma} |w_{s,\gamma}(x_0)|^2\right)^{1/2}.$$

If $K : X \to X$ is compact, then $E_{x_0}KJ$ is compact and

$$\|E_{x_0}(\mathcal{L}_s^{\mathrm{Mp}} - K)J\| \leq \|E_{x_0}\|\,\|J\|\,\|\mathcal{L}_s^{\mathrm{Mp}} - K\|.$$

Taking the infimum over compact K proves (27). Every coefficient is nonzero, so the lower bound is strictly positive. □

The theorem uses no holomorphic branch localizer. It applies to a literal bounded realization on a point-evaluative Hardy, Bergman, H^∞, or reproducing-kernel space whenever constants belong continuously. It does not cover a non-tensor anisotropic distribution space for which evaluation or every fibre slice is unbounded. This explicit hypothesis boundary is consistent with general vector-valued composition theory, where compactness depends on the operator weight and not only on the base map [4].

Theorem 5.2 (Same-domain dichotomy)**.** *On*

$$\mathcal{H}_{\mathscr{F}} = A^2(\Omega;\mathscr{F}) \simeq A^2(\Omega)\widehat{\otimes}\mathscr{F},$$

the series in (26) *converges absolutely in operator norm for every* $s \in \{s \in \mathbb{C} : \Re s > -\sigma_0\}$*, but its sum is noncompact. Thus*

$$\boxed{L_s : A^2(\Omega) \to A^2(\Omega) \text{ is trace class}, \qquad \mathcal{L}_s^{\mathrm{Mp}} : \mathcal{H}_{\mathscr{F}} \to \mathcal{H}_{\mathscr{F}} \text{ is noncompact.}} \tag{29}$$

Proof. Let $K_P = p_P(\overline{\mathcal{D}}) \Subset \Omega$. Point evaluations on scalar and vector-valued Bergman space are uniformly bounded on K_P. Since every branch image lies in K_P,

$$\|w_{s,\gamma}U_\gamma(F \circ h_\gamma)\|_{A^2} \leq C_{\Omega,P}\|w_{s,\gamma}\|_\infty\|F\|_{A^2}.$$

Unitarity removes U_γ from this norm estimate, and Lemma 3.1 makes the operator series absolutely convergent. Constants and interior evaluation are bounded on Bergman space, so Theorem 5.1 proves noncompactness. The scalar assertion is Theorem 3.2. □

For the usual volume normalization,

$$\|J\| = \mathrm{vol}(\Omega)^{1/2}, \qquad \|E_{x_0}\| = K_\Omega(x_0,x_0)^{1/2},$$

where K_Ω is the scalar Bergman kernel. The obstruction is therefore quantitative on the domain that supports the scalar determinant.

The conclusion concerns ordinary compact and nuclear operator theory. The character of the infinite-dimensional Weil representation is a distributional representation character, with a regular-locus expression, not the Hilbert trace of an isolated unitary [8]. Likewise, the oscillator semigroup studied by Hilgert [7] imposes analytic semigroup geometry in the fibre. Complexifying the AGY base does not place its real symplectic cocycle in that semigroup. A Bargmann transform only unitarily changes the oscillator model and cannot remove (27). Heat damping would change the candidate unless it were forced by a separately proved cocycle law.

Noncompactness excludes trace class, every finite Schatten class, and the ordinary nuclear Fredholm determinant for $\mathcal{L}_s^{\mathrm{Mp}}$. It does not assert that $I - u\mathcal{L}_s^{\mathrm{Mp}}$ is never Fredholm, and it does not exclude a separately constructed flat, distributional, or semifinite determinant. Most importantly, no factor such as $\operatorname{tr} U_{\boldsymbol{\beta}}$ may be inserted into (19): that ordinary trace does not exist, and the metaplectic lift sign is invisible to $\chi_{\boldsymbol{\beta}}$.

6 Exact certificate and independent replay

The infinite theorems above do not depend on a periodic cutoff. A finite certificate nevertheless fixes the source section, catches chronology and dimension mutations, and supplies a rational essential-norm witness. The selected neat strongly positive loop is

$$\gamma_* = t^{64}(tbttbtbb)^8, \qquad |\gamma_*| = 128. \tag{30}$$

Later elementary arrows multiply on the left, giving

$$B_{\gamma_*} = \begin{pmatrix} 18540 & 1210580 & 11430 & 27373 \\ 24020 & 1568410 & 14783 & 35450 \\ 50233 & 3279928 & 31130 & 74253 \\ 38803 & 2533625 & 24020 & 57343 \end{pmatrix}, \qquad \det B_{\gamma_*} = 1. \tag{31}$$

All entries are positive. Put $R_{\gamma_*} = B_{\gamma_*}^T$ and

$$x_0 = \frac{(131596, 8592543, 81363, 194419)}{8999921}. \tag{32}$$

The exact normalizer and projective Jacobian are

$$S_*(x_0) = \mathbf{1}^T R_{\gamma_*} x_0 = \frac{15076979616018}{8999921}, \tag{33}$$

$$J_{\gamma_*}(x_0) = S_*(x_0)^{-4} = \frac{6560769639033108250634950081}{51672252134321473356696529937672668896230786946152976}. \tag{34}$$

The exponent four in (34) is the homogeneous projective-Jacobian exponent, not the affine dimension three.

Keeping only this atom in (27) gives

$$\|\mathcal{L}_s^{\mathrm{Mp}}\|_{\mathrm{ess}} \geq \frac{S_*(x_0)^{-(\Re s+4)}}{\|E_{x_0}\| \, \|J\|}. \tag{35}$$

At $\Re s = 0$, the numerator is exactly (34). Its numerical size, approximately $1.2696891 \times 10^{-25}$, is irrelevant: strict positivity is enough to exclude compactness.

The positive-prefix geometry also has exact sentinels. Normalizing the columns of $P = R_{\gamma_*}$ gives

$$\delta_P = \min_{i,j} \frac{P_{ij}}{\sum_k P_{kj}} = \frac{14783}{1642663} > 0. \tag{36}$$

The maximum Birkhoff cross ratio is

$$\theta(P) = \frac{12206150825}{12121793906}, \qquad \tanh\left(\frac{\log\theta(P)}{4}\right) \approx 0.0017337504976364321. \tag{37}$$

These numbers are not needed by the complex-cone proof, but they independently verify strict positivity, precompactness, and a strong real contraction margin.

For compact notation below, set $G = \gamma_*$, $R_2 = G\, bttbtbb\, G$, and $R_3 = G\, bbb\, G$.

For the second row the forward matrix and operator-factor order are

$$B_2 = B_{R_2} B_G, \qquad (R_2, G).$$

Reversal changes this matrix, but not its characteristic polynomial: the two invertible products $B_{R_2} B_G$ and $B_G B_{R_2}$ are similar. Thus this is a contravariant matrix-order sentinel, not a spectral

Table 2: Chronological Perron data. The reciprocal polynomial is $t^4 - at^3 + bt^2 - at + 1$. Decimal Perron roots are displays of exact integer matrices; the theorem is the symbolic identity in Theorem 4.1.

Forward return word	Exact and numerical Perron data
G	Length 128; $a = 1{,}675{,}423$; $b = 463{,}448{,}097$. $\lambda \approx 1{,}675{,}146.338740454$; relative check error 5.4×10^{-95}.
G, then R_2	Length 391; $a = 16{,}015{,}896{,}888{,}538{,}880{,}980$. $b = 540{,}332{,}039{,}590{,}143{,}109{,}406{,}685{,}478$. $\lambda \approx 1.6015896888505144\times10^{19}$; relative check error 6.9×10^{-103}.
G, then R_2, then R_3	Length 650; $a = 78{,}462{,}677{,}068{,}799{,}478{,}932{,}275{,}282{,}109{,}777$. $b = 193{,}788{,}302{,}599{,}840{,}312{,}521{,}254{,}027{,}654{,}987{,}461{,}135{,}596{,}759$. $\lambda \approx 7.8462677068799479\times10^{31}$; relative check error 1.0×10^{-110}.

Table 3: Exact result ledger. "Inherited" identifies inputs proved in the preceding project stages; the present paper does not relabel them as new C26 theorems.

Statement	Status	Evidence and scope
Common complex domain	Proved here	Fixed positive prefix and Lemma 2.1 and theorem 2.2
Complex ℓ^1 branch sum	Proved here from AGY inputs	One principal logarithm plus the real exponential-tail branch sum
Scalar trace class	Proved here	Bandtlow–Jenkinson hypotheses verified in complex dimension three
Perron trace atom	Proved here	Exact telescoping and quotient derivative; three finite replays are checks only
Branch-matrix injectivity	Inherited	All-length fixed-start decoder; no finite cutoff is used as proof
Discrete atom separation	Inherited	Coherent-state essential-norm theorem, restated in Lemma B.1
Twisted Bergman noncompactness	Proved here	Same-domain evaluation slice and inherited inputs
Exact source witness	Independently verified	Separate producer/checker, 14 checks and 21 regression tests
Hilbert–Pólya bridge	Not established	No self-adjoint operator or Riemann divisor comparison

one. The third row is noncyclic: its forward matrix is $B_3 = B_{R_3}B_{R_2}B_G$, with operator factors (R_3, R_2, G). Reversing forward order gives $B_G B_{R_2} B_{R_3}$, whose reciprocal coefficients are exactly

$$a_{\text{rev}} = 78{,}462{,}676{,}839{,}072{,}961{,}074{,}051{,}275{,}858{,}065,$$
$$b_{\text{rev}} = 193{,}788{,}302{,}599{,}985{,}746{,}036{,}399{,}604{,}537{,}872{,}487{,}742{,}569{,}559.$$

They differ from the third-row (a, b), so the three-return example is a genuine spectral chronology sentinel. A centered high-precision finite difference of the three-dimensional projective derivative agrees with $\chi'(\lambda)/\lambda^3$ to the errors shown in the table.

The producer and checker independently reconstruct seven Rauzy states, fourteen edges, the complete 128-arrow word, all three Perron examples in Table 2, and every rational value above. The checker does not import the producer and uses a different derivative computation. A separate length-20 mutation sentinel finds 13,528 first returns and 13,528 distinct matrices. That finite replay tests code; it is not evidence for all-length injectivity or completeness. Full commands and artifact hashes are recorded in Appendix C.

Table 4: Route-A evaluation of the targeted metaplectic candidate. The scalar determinant is reusable infrastructure but does not change the target verdict.

Gate	Verdict	Strongest evidence	Decisive failure
A1: arithmetic orbit structure	Weak	Chronological matrices are all-length decodable; trace atoms use algebraic Perron units	No prime-like primitive law or logarithmic-prime clock
A2: operator/determinant	Fail	Scalar L_s is trace class on the common domain	Targeted oscillator operator has positive essential norm
A3: zeta structure	Fail	$D(s,u)$ is an ordinary scalar Fredholm determinant	No functional equation, gamma factor, counting law, or Riemann divisor
A4: Hilbert–Pólya bridge	Formal hint	Exact symplectic cocycle and natural Weil representations	No surviving self-adjoint operator or bridge from its spectrum to ξ

7 Hilbert–Pólya assessment and the next large door

Table 4 evaluates the literal source-locked object: AGY return clock, raw Jacobian weight, chronological cocycle, and unsmoothed oscillator representation. It fails before a zero comparison is legal; no zero fit, prime table, Hermite cutoff, or heat factor enters the test.

The outcome is

$$(\text{A1\_WEAK}, \text{A2\_FAIL}, \text{A3\_FAIL}, \text{A4\_FORMAL\_HINT}), \qquad \text{ROUTE A REJECTED}.$$

The result closes the direct holomorphic escape on standard point-evaluative tensor spaces; non-tensor spaces without bounded slices remain outside it. The regular-locus Weil character of Thomas [8] is distributional, not the ordinary trace in (26). The oscillator semigroup of Hilgert [7] contracts analytically, but the real AGY cocycle does not automatically lie in it. Either move needs a new source lock.

The next large gate changes the fibre while preserving base and chronology. For odd p, reduce the integral symplectic matrices and use

$$g_\gamma \bmod p \in \mathrm{Sp}(4, \mathbb{F}_p), \qquad \rho_p(g_\gamma) \in U(p^2). \tag{38}$$

Finite-field quantization supplies genuine characters [6]. This fibre preserves scalar trace class and chronology while avoiding infinite multiplicity; it is a new candidate, not a regularization or a claim about $p \to \infty$.

The first experiment should compute exact character weights at several odd primes and test conjugacy, central lifts, and symbolic primitive classes. Even a positive signal would still need a functional equation, counting law, and self-adjoint bridge before Route B.

8 Conclusion

On one source-faithful AGY complex domain, the fixed positive prefix gives the scalar operator trace class, an ordinary Fredholm determinant, and atom $\lambda^{-(s+1)}/\chi'(\lambda)$. Constants and evaluation instead give the literal oscillator twist positive essential norm throughout $\Re s > -\sigma_0$. This excludes its ordinary nuclear determinant on the stated tensor spaces, not distributional or non-tensor theories.

No Hilbert–Pólya conclusion follows: there is no self-adjoint target, functional equation, counting law, or Riemann-zero match. The next large test is the finite Weil pivot in (38), changing the failed fibre while retaining exact chronology and scalar trace class.

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A Complex-domain audits

A.1 Why the natural image domain is too small

Let Δ be the real AGY base inside $p_P(\mathcal{D})$. The inverse branch images $h_\gamma(\Delta)$ form the full-branch partition of Δ modulo a null set. Suppose that one compact set $K \Subset p_P(\mathcal{D})$ contained every complex branch image of that same domain. Its real intersection would contain every $h_\gamma(\Delta)$. Since $K \cap \Delta$ stays a positive distance from the relative boundary, it cannot cover the positive-measure open region of Δ near that boundary. This contradicts the full-branch partition. Therefore one must enlarge past $\overline{p_P(\mathcal{D})}$, as in (6); uniform compact containment on the natural domain itself is false.

A.2 Real contraction need not complexify uniformly

The fixed positive prefix cannot be replaced by an abstract appeal to real Hilbert contraction. Let $\varepsilon_n \downarrow 0$ and consider

$$f_n(z) = \frac{1}{2} + \frac{1}{4}\left(z - \frac{1}{2}\right) + \frac{\varepsilon_n^4}{(z - \frac{1}{2})^2 + \varepsilon_n^2}.$$

For real $z \in (0,1)$, the last term lies in $(0, \varepsilon_n^2]$, and its derivative has magnitude $O(\varepsilon_n)$. After discarding finitely many indices, all f_n map $(0,1)$ into one fixed compact subinterval and have derivative bounded by a constant strictly below one. They are therefore uniformly contractive for both Euclidean and Hilbert metrics on a fixed real subinterval. Nevertheless, f_n has poles at $1/2 \pm i\varepsilon_n$. No common complex neighborhood of $(0,1)$ supports the full family. In Theorem 2.2, the nonnegative matrix factorization prevents this pole degeneration.

A.3 Boundary-degenerate normalized matrices

For numerical estimates one may divide a branch matrix by its total entry sum,

$$\widehat{A}_\gamma = \frac{A_\gamma}{\sum_{i,j}(A_\gamma)_{ij}}.$$

The normalized family lies in the compact simplex of nonnegative matrices. Its limits may lose columns or rank. This does not threaten the common denominator: if $x_j \geq \delta > 0$ on a real compact set, then

$$\ell(\widehat{A}_\gamma x) = \sum_j \left(\sum_i \widehat{A}_{\gamma,ij} \right) x_j \geq \delta.$$

The fixed factor $P > 0$ prevents projective outputs from approaching the boundary of the full positive cone. This calculation explains why the exact coordinate margin (36) is a useful mutation sentinel. It does not authorize replacing the raw integral matrix in the roof or trace formula: scaling A_γ changes $q_\gamma^{-(s+d)}$.

A.4 Complex powers and word telescoping

The algebraic identity

$$q_A(z)q_B(p_A z) = q_{BA}(z)$$

does not by itself justify multiplication of nonintegral complex powers on an arbitrary domain. On $\Omega \Subset \mathcal{D}$, every partial composite has positive real denominator, so the principal logarithms are compatible. Equivalently, the two holomorphic logarithmic expressions agree on the real positive simplex and hence throughout the connected domain. The proof of Theorem 4.1 uses the still safer periodic-point argument: every intermediate normalizer is positive real, and its product is exactly the Perron eigenvalue.

A.5 An equivalent tubular construction

The complex-cone domain is the cleanest global construction. A local audit can instead complexify the Hilbert Finsler norm on $K = \overline{\Delta} \Subset \{x_i > 0\}$. Uniform real contraction and the total-entry normalization above give a Taylor estimate

$$\rho(h_\gamma z) \leq q\rho(z) + C\rho(z)^2, \qquad q < 1,$$

where ρ is the distance to K in a uniformly equivalent family of complexified Hilbert norms. A sufficiently small sublevel set then maps into a strictly smaller one. This alternative proves existence but hides the stronger algebraic fact in (7): the entire common image is forced by P, independently of branchwise Taylor constants.

B Operator slices and inherited inputs

This appendix restates the two inherited project inputs in the precise form used by Theorem 5.1. The first is an oscillator essential-norm theorem. The second is fixed-start injectivity of the full Rauzy matrix. Neither finite certificate in Section 6 is used as a substitute for these all-length statements.

Lemma B.1 (Discrete metaplectic atoms)**.** *Let g_k be pairwise distinct elements of* $\mathrm{Sp}(2m,\mathbb{R})$*, choose a lift $\widetilde{g}_k \in \mathrm{Mp}(2m,\mathbb{R})$ for each, and suppose $(a_k) \in \ell^1$. On $L^2(\mathbb{R}^m)$,*

$$V = \sum_k a_k \mu(\widetilde{g}_k)$$

satisfies

$$\|V\|_{\mathrm{ess}} \geq \left(\sum_k |a_k|^2\right)^{1/2}.$$

If several lifts project to the same g, one must first combine their coefficients with the actual central signs.

Proof. Let $W(z)$ be the Weyl representation and choose a unit Schwartz vector ϕ. Select $z \neq 0$ outside the countable union of proper subspaces $\ker(g_k - g_l)$, $k \neq l$. The coherent states $\phi_t = W(tz)\phi$ converge weakly to zero. Metaplectic covariance gives, up to phases,

$$\mu(\widetilde{g}_k)\phi_t = W(tg_k z)\mu(\widetilde{g}_k)\phi.$$

Weyl matrix coefficients of Schwartz vectors vanish when their phase-space separation tends to infinity [5]. Hence all cross terms between $k \neq l$ vanish as $t \to \infty$, while diagonal terms have norm one. Since $(a_k) \in \ell^1$, dominated convergence yields

$$\|V\phi_t\|^2 \longrightarrow \sum_k |a_k|^2.$$

Every compact operator sends ϕ_t to zero in norm. Taking the distance from compact operators proves the estimate. Equal projected atoms must be aggregated first because the two metaplectic lifts differ by a central sign. □

B.1 Fixed-start matrix injectivity

For a labeled Rauzy path starting at a fixed permutation, its full chronological matrix determines its first arrow. The two outgoing arrows exchange winner and loser; for a valid image matrix, exactly one of the two candidate winner-row subtractions remains componentwise nonnegative. Subtracting that loser row strictly decreases the sum of all entries and recovers the remaining path matrix. Induction terminates at the identity and recovers the full word. Thus distinct fixed-start return paths have distinct matrices. In the present full-rank $\mathcal{H}(2)$ model, one fixed symplectic frame conjugates every closed branch matrix to its projected symplectic action, so projected matrices are distinct as well. The declared edge lifts then recover, rather than average, the pathwise central sign.

B.2 General tensor slices

Point evaluation is one instance of a broader obstruction. Let $I_f : \mathscr{F} \to X_0$ and $R_\ell : X_1 \to \mathscr{F}$ be bounded maps such that

$$R_\ell \mathcal{T} I_f = \sum_\gamma b_\gamma U_\gamma, \qquad (b_\gamma) \in \ell^1.$$

After aggregating equal projected atoms with their central signs, Lemma B.1 and the ideal property of compact operators give

$$\|\mathcal{T}\|_{\mathrm{ess}} \geq \frac{\|(b_g)\|_{\ell^2}}{\|R_\ell\|\,\|I_f\|}.$$

This version covers tensor-type anisotropic spaces only when the proposed input and output slices are actually bounded. It makes no assertion about a completion designed to correlate base and oscillator modes so strongly that every such slice fails.

B.3 Why the scalar trace cannot be twisted formally

For an infinite-dimensional unitary U, one has $|U| = I$, so $\operatorname{tr}|U| = \infty$. Even if a scalar branch operator is trace class, its tensor product with U has every nonzero singular value with infinite multiplicity. The distribution character in Thomas [8] becomes a function only on a regular locus after the representation is treated in its appropriate distributional sense. It is not an ordinary factor that can be multiplied into (19). A finite-dimensional fibre would be different: its chronological matrix trace is legal and preserves scalar trace class.

C Reproducibility and disclosure

C.1 Reproduction contract

From the repository root, regenerate and validate the exact certificate by running

```
cd henon_dynamics/agy_holomorphic_slice_obstruction
./code/run_c26.sh
```

The runner executes the exact producer, an independent checker, the regression and mutation suite, and artifact-hash verification. The released run used Python 3.12.3 and SymPy 1.14.0. It reports seven Rauzy states, fourteen directed edges, a length-128 source word, fourteen independent checks, and twenty-one tests. The checker imports neither producer functions nor producer constants. It reconstructs every elementary Rauzy move and uses a centered finite difference rather than the producer's analytic projective derivative.

The two principal JSON artifacts have the recorded SHA-256 digests

$$\begin{aligned} \texttt{c26\_certificate.json}: &\quad \texttt{1c0289b9b47e65e0603ea001be7cce263}\ldots, \\ \texttt{c26\_independent\_check.json}: &\quad \texttt{20cddc20923184de687b8e70ee3bf7b4d}\ldots. \end{aligned}$$

The complete values are stored in `results/ARTIFACT_HASHES.sha256`; abbreviated hashes above are for readability, not for verification.

The mutation suite rejects reversed chronology, B/B^T confusion, Jacobian exponents three or five, complex dimension four, a zero entry in the positive prefix, a branchwise fitted logarithm, the two-return contravariant matrix order, and the three-return noncyclic spectral order. It also rejects barycentric substitution for x_0, omission of either bounded slice, and cancellation errors among equal projected atoms. The finite length-20 decoder replay is registered explicitly as a non-proof sentinel.

Compile this article with

```
cd paper
latexmk -pdf -interaction=nonstopmode -halt-on-error main.tex
```

The source is modular: every main section and appendix is a separate file, and the bibliography contains only cited, DOI-verified references.

C.2 Evidence levels

The common-domain, scalar trace-class, Perron trace, and evaluation-slice arguments are mathematical proofs. The exact scripts verify the selected section word and three Perron specializations; they do not enumerate the countable AGY family or prove a theorem by finite search. The all-length decoder and discrete-atom theorem are inherited mathematical inputs, restated in Appendix B. The finite-Weil direction in Section 7 is a proposed experiment, not a result.

C.3 AI-assistance disclosure

AI systems assisted with symbolic exploration, source organization, independent code generation, theorem auditing, and preparation of the manuscript. Every cited bibliographic record was checked against a DOI, article record, or publisher catalog. Exact producer outputs were replayed by separately written checker code, and claims were assigned explicit evidence levels before inclusion. AI assistance does not replace the mathematical arguments: the paper states all hypotheses used by the infinite-family theorems, distinguishes inherited results from present deductions, and records negative conclusions and open boundaries without upgrading them to proofs.